

## Research article

# The impact of informality and institutional quality on environmental footprint: The case of emerging economies in a comparative approach

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## ABSTRACT

The endeavor to implement the 2030 Agenda of national and international stakeholders became increasingly impetuous, considering the wide range of uncertainties and risks. The new humans-centered development model built on the prominence of environmental and social values seeks to reinforce communities' resilience and mitigate environmental risks, leaving no one behind. For this to happen, solid and effective institutions, the right environmental policies, and a safe statutory labor framework are the sine qua non. In this study, we evaluated the effects of informality, institutional quality, and renewable energy consumption on ecological footprint of two groups of emerging countries from Europe and Asia from 2002 to 2018. Our results by PMG-ARDL approach highlight dissimilarities between the two groups, showing greater heterogeneity. In this sense, informality is found to have positive and significant effects on ecological footprint in the long run in emerging European countries. In contrast, the effect is negative for emerging Asian countries. In the short run, the effects are less critical in the studied countries between the two groups. Institutional quality variables impacted environmental quality differently. In this sense, it is detrimental for policymakers to consider concerted measures to decrease institutional vulnerabilities and reduce the level of the informal economy. The outcome of this study concurs with a complete awareness of the importance of informality and institutional quality to mitigate social and environmental risks conjunctively.

## 1. Introduction

Emerging countries, in their swift development, are marked by a prevailing level of informality, which is upheld by favorable macro-economic factors (Bolarinwa and Simatele, 2023; Elgin et al., 2021; Medina and Schneider, 2018; Medina and Schneider, 2019; Sánchez-Vidal et al., 2020; Wang et al., 2021), such as the institutional framework, regulatory capacity, poverty, tax burden, structural factors (Canh et al., 2021; Dreher et al., 2009), economic complexity (Nguyen, 2022), past levels of inequality (Gutierrez-Romero, 2020) financial and investment climates (Canh and Thanh, 2020; Huynh et al., 2020), tax evasion (Dell'Anno and Davidescu, 2019; Torgler and Schneider, 2009) or the level of qualification of the workers (Delechât and Medina, 2020). In addition, a high level of environmental damage is observed in emerging markets and developing economies (EMDEs). As suggested by

IEA (2021), globally, EMDEs account for two-thirds of global carbon emissions, mainly due to coal-fueled energy, while they are accounting for almost 50% of the global economic growth (decreasing from 70% in 2017) (World Bank Group, 2023), experiencing a slow growth trend due to the negative impact of multiple factors (health crisis, geopolitical tensions, higher inflation, energy crisis, climate change challenges). The quality of environment is closely associated with the effectiveness of regulating this field, and securing compliance with these regulations is a goal, both at the microeconomic and macroeconomic level. Additionally, the lack of official data impedes setting strict rules and making accurate decisions. The intensive industrial process specific to emerging economies and the attitude towards these regulations, in the broader context of rapid economic development objectives, is closely linked to structural and institutional factors, which also determine the level of the informal sector. In this context, political actions are needed to target

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these two phenomena, which are correlated at a certain level, and informality must be analyzed as a factor associated with environmental quality (Hunjra et al., 2023a; Mitoma, 2023; Sultana et al., 2022a; Wang et al., 2019).

Considering the definitions and interpretations of the informality concept in the economic literature (Lührmann, 2010; Medina and Schneider, 2019; Williams and Horodnic, 2016; Williams and Öz-Yalaman, 2021) and the difficulty of reporting and highlighting the level of informality (ILO, 2023), we want to emphasize our point of view in this respect. Namely, informality considers legal actions and activities carried out outside the natural legal framework, especially for the consideration of saving the part of income that would have been automatically transferred in the formal framework to the state institutions to ensure and protect the interests of workers or employers, also in line with González-Fernández and González-Velasco (2015) view. The compliance of the potential taxpayers (individuals or firms) is strongly related to their perceptions about the state institutions' legitimacy. The lack of trust in the social contract partner, the continuous postponement of ensuring the decent conditions necessary for daily living, and the depersonalization of state institutions in relations with individuals/workers affect the cost-gain balances and may diminish the non-ethical aspect of hiding parts of revenues from taxation (Tanzi, 2002; Torgler and Schneider, 2009). Additionally, this behavior is also related to a short-term benefits approach (in the short term, the individual has a larger amount of money if he does not pay taxes and social security contributions), forsaking the potential long-term benefits. This short-term view specific to informality is also visible in environmental degradation process.

The informality phenomenon becomes increasingly complex as it is analyzed, being both a supply and demand feature (Bolarinwa and Simatele, 2023). Although, it occupies an important part of the economy, it is difficult to measure and appreciate its magnitude and impact on environmental degradation. Both society and the individual perceive informality in a dual perspective. First, it is perceived as the only way to obtain (additional) revenue in a society where economic disparities are challenging to flatten. Secondly, it is seen as violating the fiscal and working legal framework imposed by the state authorities. Informality is a phenomenon that slowly reduces its size as much as the economies are developing, but differently from one region to another, remaining at about a third of low- and middle-income countries' economic activity and 15 percent in advanced economies (Delechat and Medina, 2020). Emerging economies from Europe also have higher shares of the shadow economy, on average around 30–35 percent of GDP.

It also should be noted that the presence of informality can be perceived as a crowding out effect for the ability of governments to support the new green, resilient, and inclusive development model; in this sense, the higher the level of informality, the lower will be the financial and decision-making capacity of governments in supporting sustainable development of nations. Moreover, informality is an effect of underdevelopment, and specific solutions must be found for each country because the causes that generate the level of the shadow economy are different (Ohnsorge and Yu, 2022). Although the quality of the environment represents an extensive concern in the literature (Abban et al., 2023; Alkathery et al., 2022; Alkathery et al., 2023; Ayad et al., 2023; Cutcu et al., 2022; Cutcu et al., 2023; Fayyaz et al., 2023; Hunjra et al., 2023b; Islam et al., 2023; Karim et al., 2023; Khan et al., 2023; Li et al., 2022; Li et al., 2023; Naeem et al., 2023; Sharif et al., 2023; Su et al., 2023; Sun et al., 2023; Ullah et al., 2023; Zameer et al., 2023), few studies highlight how informality and the quality of government institutions can influence carbon emissions, and even fewer evaluate the impact of informality on ecological footprint (Appiah et al., 2023; Sultana et al., 2022a). Ecological footprint, conceived in 1990 (Wackernagel and Galli, 2007), used in this study is a broad and wide-ranging indicator of environmental degradation (Arshad et al., 2020; Khan et al., 2022; Wang et al., 2020), calculating the ecological assets necessary for a population to produce the natural resources it

consumes (Global Footprint Network, 2022), using six indicators: carbon footprint, built-in land, forest, cropland and pasture, and fisheries measured in a consolidated unit (gha) (Akhtar et al., 2022). At the same time, due to its exhaustive assessment of environment, ecological footprint developed an essential interest in existing literature.

This study focuses on the informality and institutional indicators regarding government effectiveness, regulatory quality, the rule of law, and control of corruption on two panels of emerging countries from Europe (EEC) and Asia (AEC), using PMG-ARDL approach. We considered these two sets of countries to establish some insights starting from the characteristics of the countries and regions we choose. This analysis contributes to the literature in many aspects, considering the gaps in the reviewed literature. It focuses on selected emerging countries that are more sensitive to the informal activities of workers and firms, considering as an indicator the level of informal activity as a percent of total output; thus, it creates space to include some general recommendations, including by taking into account the specific situation of countries from the panel and different sectors. Additionally, it considers ecological footprint as a broader image of environmental degradation and evaluates the significance of informal economy level for environmental quality. Moreover, it points out institutions' quality and capacity to create a safe framework for economic activity and responsibility. For controlling, renewable energy consumption has been chosen to complete the image of the analysis. Gross final energy consumption from renewable sources (Eurostat, 2023) (biomass, hydro, wind, heat pumps, liquid biofuels, etc.) in the European Union for 2021 (21.8%) was below the target in place at this moment (32%) and too far from the new proposed target by the REPowerEU plan in 2022 (45%). Analyzing the countries proposed by this work, only Latvia (42.11%), Slovenia (25.00%), and Romania (23.59%) are above the EU average. Meanwhile, the Czech Republic (17.67%), Slovakia (17.41%), Bulgaria (17.02%), and Hungary (14.11%) are well below the target and must increase their efforts to ensure the right path. For the Asian emerging countries, the share of renewable energy consumption in 2020 are the following: Pakistan (46.60%), India, (35.82%), Philippines (29.06%), Indonesia (22.01%), Thailand (20.83%), and Malaysia (5.84%), according with the latest data (2020) from World Development Indicators (WDI). The largest share of renewable energy consumption is observed in Pakistan, with enormous potential for generating renewable energy from solar and wind power.

This study is organized as follows: Section-II presents a review of literature that assessed the influence of informality, institutional quality, and renewable energy consumption on ecological footprint. Section-III outlines the dataset, Section-IV presents the methodology, followed by results and discussion section, whereas Section-VI highlights conclusions and policy implications.

## 2. Literature review

### 2.1. Ecological footprint and informality

The informal economy can influence the quality of environment in several directions, according to the literature, and this fact creates real challenges for establishing and complying with environmental standards, especially in developing countries (Croitoru and Sarraf, 2012). Moreover, certain studies have highlighted a direct correlation between the level of informality and the level of pollution in the broader context of corruption that dilutes environmental regulations and reduces their positive effects on the quality of environment, increasing polluting emissions (Chen et al., 2018). In a very comprehensive study (Ulysea, 2020), the author concludes that informality is both a micro phenomenon (representing the effect of the individual's decision to pay less or nothing in the form of taxes or social security) and a macro phenomenon (referring to the economic and social effects it has for a specific community and to the measures of a political nature that must be approached to diminish its amplitude). To a certain extent, it could be

limited by reducing the costs of entering the formal sector and by intensifying enforcement. Also, the literature (Webb et al., 2020) explained the reasons why an employer engages with the informal economy: competition, financial constraints, lack of awareness and inappropriate understanding of compliance or a deliberate choice not to comply, or to avoid regulatory responsibilities (Medina and Schneider, 2018; Webb et al., 2019, 2020; Williams, 2011). On other hand, other studies (Shafiq, 2015) have developed analyses concluding that economic development coincided with a decline in moral values. Also, Sultana et al. (2022b) analyzed the informal sector's effect on sustainable development in 50 developing countries from 2010 to 2019 and found that the informal sector harms the selected countries' sustainable development.

Thereby, both indicators, informality (Liu et al., 2021) and environmental quality, are important factors that should be taken into consideration also conjunctively in the literature. The specific deregulation of firms activating in the informal economy represents a critical impediment to the state's actions in mitigating environmental degradation. Also, the literature discusses the spatial spillover effects of the corrupted behavior of those acting in shadow economy on neighboring countries in the context of economic and social interaction and globalization (Feng et al., 2020; Goel and Saunoris, 2020). This evidence is complemented by (Ortiz et al., 2022), who highlighted in their study on 101 countries during 1995–2018 the importance of reducing informal and clandestine economic activities as tools to mitigate environmental degradation. Considering the possible link between pollution and informality, it is essential to envisage that new requirements for strict environmental regulation can lead to adverse reactions from companies, which will tend to transfer part of their activity to the informal economy under the impact of high environmental protection costs (Pang et al., 2021).

Contrariwise, other studies consider that the shadow sector is less energy-but labor-intensive. In the short run, an increase in the unrecorded economy is linked to a decline in energy consumption (Benkraiem et al., 2019; Chen et al., 2021). Assuming this reasoning, the level of informality in a country generates less pollution than the statutory one. Pang et al. (2021) studied the interactive relationship between the shadow economy and environmental pollution (especially air pollutants) in 31 provinces in China from 2005 to 2015. Their study employed the Generalized Method of Moments in the presence of environmental regulation, economic growth, foreign direct investment, and educational level showing a positive interaction between the shadow economy and the pollution level.

In a comparative formal-informal approach of 25 manufacturing industries from India (Mitoma, 2023) pointed out the significant contribution of the informal sector for several industries, such as other non-metallic mineral products and basic iron and steel (brick production for construction) to the national carbon footprint. The authors recommended a reduction of the informal sector to improve India's environmental quality and a greener supply chain engagement between formal and informal sectors. In a study that evaluates the shadow economy's influence on forest land's environmental quality on a panel of 148 countries (Nguyen and Nguyen, 2023) found a positive impact in the short run, but with strong implications in the long run, on deforestation. Also, Bento et al. (2018) demonstrated that the presence of the informal sector would reduce the cost of environmental and energy tax policy, with more significant effect in developing countries, describing the existence of two mechanisms supporting the effect. Additionally, Lee et al. (2022) suggested that policymakers should embolden different formal and informal environmental regulation to reduce distinct types of environmental degradation.

### 3. Ecological footprint and institutional quality

The quality of public institutions can be the foundation of an environment-friendly attitude and, in general, of compliance with

regulations and statutory requirements. The Worldwide Governance Indicators (WGI) assesses the level of governance in over 200 countries (WGI, 2022). In our paper, we used four of the six dimensions of governance to determine how the government's ability to formulate and implement policies to support the private sector, the public's perception of the public policies' purpose, the extent to which the rules of society are respected, the quality of public services and independence from miscellaneous pressures. Providing a stable framework for development, institutional quality can ensure a pro-complying rules attitude, including those related to environment. A reduced level of corruption, a higher level of decision-making transparency, and an efficiency of more intense government policies constitute a good framework for organizing economic, civic, and social activities following the regulations and outside the tendency to move towards informal economy. Salman et al. (2019) found that it is very important that public institutions in East Asian countries analyzed over 1990–2016 to be impartial and independent to lessen carbon emissions and increase economic growth. In addition, a study on the Economic Community of West African States from 1990 to 2016 (Ali et al., 2022) found that the governance indicators and renewable energy consumption diminishes ecological footprint. In a study by Hussain and Dogan (2021) on BRICS countries from 1992 to 2016, they highlighted the negative effect of institutional quality on ecological footprint, advocating for supporting investments in environmental technologies for a sustainable future. Moreover, corruption affected negatively the ecological efficiency and resources allocations in China (Wang et al., 2020).

Similarly, Sinha et al. (2019) ascertained the role of corruption in heightening carbon emissions in a study on BRICS and the Next 11 countries from 1990 to 2017, highlighting the importance of government effectiveness and efficiency in setting the right policies to mitigate environmental pollution. The specific measures to control corruption and improve governance lessen the negative impact of economic growth on environment (Li et al., 2023), as has been found on a panel of 158 countries. Evaluating the corruption impact on environmental quality in 17 MENA countries from 1984 to 2018, using the method of moments quantile regression (Khalfaoui et al., 2023), found different results specific to environmental degradation. In this sense, corruption deteriorates the quality of environment more intensely at reduced levels of environmental degradation and less intensely at higher levels. Another study by Wang et al. (2019) presenting the impact of corruption and informal economy interaction on environmental quality used spatial econometrics in China from 1998 to 2017. Their study concluded that the interaction effect increases pollution significantly and recommends the control of both corruption and hidden economy to improve environmental quality. This evidence about the potential risk of corruption in accomplishing sustainable development goals is complemented by Akalin et al. (2021) in a study on Newly Industrialized Countries from 1984 to 2016 using panel data approaches. They found a positive long-term relationship between corruption, fossil fuel usage, and environmental degradation. Moreover, renewable energy consumption has been found to have a negative impact on ecological footprint. The size of informality has been found to have a significant impact on local pollutants in developing countries and a negative effect on global pollutants in developed countries. Governance impact is important when it refers to controlling corruption, having a negative impact on carbon emissions, and having a positive effect on ecological footprint.

#### 3.1. Ecological footprint and renewable Energy consumption

Growing literature analyzes the impact of renewable energy consumption on carbon emissions. In this sense, energy consumption from renewable sources (Abban et al., 2022; Dilanchiev et al., 2023; Nuță et al., 2021; Zhang et al., 2023) has been considered an efficient factor in mitigating environmental degradation: (Khan et al., 2021) findings for a global panel of 188 countries during 2002–2018, in the presence of technological innovation; (Wang et al., 2020) for N-11 countries from

1990 to 2017. Also, (Nguyen and Kakinaka, 2019) found that renewable energy consumption has a positive association with carbon emissions in low-income countries but a negative association with emissions in high-income countries, studying 107 countries from 1990 to 2013. Thus, Sharif et al. (2020) and Vo et al. (2022), re-investigating the impact of renewable and non-renewable energy consumption on Turkey's ecological footprint using a Quantile Autoregressive Lagged approach during 1965–2017, found that renewable energy decreased environmental footprint in the long run on each quantile. Increasing consumption of renewable energy sources (Sharif et al., 2021; Wang et al., 2020) and complementary decreasing fossil fuels are presented in the literature as a solution to curtail environmental degradation. Sharma et al. (2021) found that intensive energy consumption from renewable sources contracted the ecological footprint in eight developing countries from South and Southeast Asia in the long run. Renewable energy has become one of the most critical and safe engines necessary for sustainable development in the new context assumed by international political and institutional leaders. Moreover, informal economy is found to be damaging on renewable energy in a study on 30 high and middle-income countries from 1990 to 2020 (Chu et al., 2023).

Using the Panel Smooth Transition Model to evaluate the nonlinear link between renewable energy consumption and ecological footprint for the highest 15 renewable consumers at the global level, (Ullah et al., 2021) found a negative sign between the two variables, confirming the positive effect of renewable energy sources in mitigating environmental risks. Considering the importance of these results, they suggest the need to intensify renewable energy in industrial and technological processes to ensure healthy economic development. Similar results were found in Italy by Javed et al. (2023). Moreover, renewable energy consumption was found to inhibit carbon emissions in a study (Balsalobre-Lorente et al., 2023) on Central and European emerging economies, being a valid method to limit environmental degradation. Also, the increase in renewable energy consumption decreases ecological footprint (Abid et al., 2022) in Saudi Arabia. These results are complemented by Pata (2021), indicated that renewable energy diminishes environmental degradation in China and Brazil, pointing out the critical role of renewable to environmental degradation mitigation. In an attempt to evaluate factors influencing ecological footprint in 103 countries, applying Necessary Condition Analysis (NCA) and Fuzzy-set Qualitative Comparative Analysis models, Kazemzadeh et al. (2023) suggest that energy consumption is a necessary condition for evaluating ecological footprint. Complementary, their results show that institutional quality, together with energy consumption, led to an improvement in the quality of environment. By contrast, despite the benefit of renewable energy for environmental protection, Appiah et al. (2023) found a positive relationship between renewable energy and ecological footprint for 29 OECD countries in a study from 1990 to 2020, highlighting that renewable from biofuels should be limited in these countries. Decision-makers from the analyzed emerging countries must engage in projects that allow them to improve government efficiency, mitigate climate change-related risks and ensure the resilience of the states in the conditions of no one left behind, also addressing gender inequality issues, knowing that a large part of the people involved in the informal economy there are women or people with low skills.

#### 4. Data

As previously stated, the main aim of this paper is to investigate the effect of informal economy and governance on environment quality in seven European emerging countries (Bulgaria, Czech Republic, Hungary, Latvia, Romania, Slovak Republic, and Slovenia) and seven Asian emerging countries (India, Indonesia, Malaysia, Pakistan, Philippines, Thailand, and Vietnam). Thus, we use annual data covering 2002–2018 for several variables from different sources (Table 1).

Thus, instead of CO<sub>2</sub> emissions focusing only on air pollution, this study use ecological footprint (measured by global hectares) to indicate

environmental quality. This indicator allows us to measure the effect of people's pressure on environmental deterioration, including CO<sub>2</sub> emissions in addition to the fishing ground, forest products, water pollution, cropland, and build-up land as can be found in the Global Footprint database. Second, the study uses Informality, measured as a share of official GDP, as a proxy of the informal output (% of official GDP) estimated by Elgin et al. (2021), using a DGE model. Third, we use renewable energy consumption as a control variable from WDI. Fourth, we used four different worldwide governance indicators (Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption) from World Development Indicators databases.

#### 4.1. Methodology

The general form of the ARDL model is given in Eq. (1).

$$Y_t = c_0 + c_1 t + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{i=0}^q \beta_i X_{t-i} + u_t \quad (1)$$

$t = \max(p, q), \dots, T$ , for simplicity, assume that the lag order  $q$  is the same for all variables in the  $K \times 1$  vector  $x_t$ .

Pooled Mean Group (PMG) enables intercepts, short-run coefficients, and error variances to differ across groups and evaluations of the long-run homogeneity relationships, Eq. (2).

$$\Delta y_{i,t} = \phi_i EC_{i,t} + \sum_{j=0}^{q-1} \Delta X_{i,t-j} \beta_{i,j} + \sum_{j=1}^{p-1} \lambda_{i,j} * \Delta y_{i,t-j} + \varepsilon_{i,t} \quad (2)$$

where  $EC_{i,t} = y_{i,t-1} - X_{i,t} \theta$ , assuming that the regressors  $X$  have the same number of lag  $q$  in each cross-section (Pesaran et al., 2001). Accordingly, the estimation model is given in Eq. (3).

$$EF_{it} = \alpha_1 + \alpha_2 INF_{it} + \alpha_3 REN_{it} + \alpha_4 GE_{it} + \alpha_5 RQ_{it} + \alpha_6 RL_{it} + \alpha_7 CC_{it} + \varepsilon_{it} \quad (3)$$

where EF is ecological footprint, INF is the informal economy; REN is renewable energy consumption. GE, RQ, RL, and CC are Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption, respectively. Additionally,  $\alpha_k$  and  $\varepsilon_{it}$  are the model parameters and the white noise of the estimations, respectively.

Pesaran and Yamagata (2008), developed a method to test the phenomenon of slope homogeneity for panel data with large  $N$  (cross-sectional dimension) and  $T$  (time dimension), assuming that  $\varepsilon_{i,t}$  and  $\varepsilon_{j,s}$  are independently distributed for  $i \neq j$  for  $t \neq s$ , allowing for a heterogeneous variance. The test statistic is described by Eq. (4).

$$\tilde{\Delta} = \frac{1}{\sqrt{N}} \left( \frac{\sum_{i=1}^N \tilde{d}_i - k_2}{\sqrt{2k_2}} \right) \quad (4)$$

where  $\tilde{d}_i$  weighted difference between the cross-sectional unit specific estimate and the pooled estimate in Eq. (5).

$$\tilde{d}_i = (\hat{\beta}_{2i} - \hat{\beta}_{2WFE}) \frac{X'_{2i} M_{1i} X_{2i}}{\hat{\sigma}_i^2} (\hat{\beta}_{2i} - \hat{\beta}_{2WFE}), \quad (5)$$

where  $X'_{2i} = (X_{2i,1}, \dots, X_{2i,T_i})'$ ,  $M_{1i} = I_{T_i} - Z_{1i}(Z'_{1i}Z_{1i})^{-1}Z'_{1i}$  and  $Z_{1i} = \tau_{T_i}$ ,  $X_{1i}$  with  $\tau_{T_i}$  being a  $T_i \times 1$  vector of unity, representing the constant and  $\hat{\beta}_{2i}$  and  $\hat{\beta}_{2WFE}$  representing coefficients.

In the case of normally distributed errors, the mean-variance bias adjusted  $\tilde{\Delta}$  are given by Eqs. (6) and (7).

$$\tilde{\Delta}_{adj} = \sqrt{N} \left( \frac{N^{-1} \sum_{i=1}^N \tilde{d}_i - k_2}{\sqrt{V_{ar}(\tilde{z}_i, T_i)}} \right) \quad (6)$$

where

$$V_{ar}(\tilde{z}_i, T_i) = \frac{2k_2(T_i - k - 1)}{T_i - k_1 + 1} \quad (7)$$



The second step in this study is to explore the cross-section dependence (CSD) issue across the countries to identify the optimal generation of unit root tests and avoid biased estimates. Hence, we use three different tests to get a clear idea about CSD such as (Breusch and Pagan, 1980) LM (BP LM) (Pesaran, 2004), Scaled LM (PS LM), and Bias corrected scaled LM test (BCS LM) presented by (Baltagi et al., 2012). The form of Breusch and Pagan (1980) LM test can be applied to test for the cross-sectional dependence in heterogeneous panels. In this case, it is given by Eq. (8).

$$BP_{LM} = T \sum_{i=1}^{n-1} \sum_{j=i+1}^n \hat{\rho}_{ij}^2 \quad (8)$$

The form of the scaled version proposed by Pesaran (2004) is given by Eq. (9).

$$PS_{LM} = \sqrt{\frac{1}{n(n-1)}} \sum_{i=1}^{n-1} \sum_{j=i+1}^n (T\hat{\rho}_{ij}^2 - 1) \quad (9)$$

The form of bias-corrected LM test statistic (Baltagi et al., 2012) is given in Eq. (10).

$$BCS_{LM} = LM_P - \frac{n}{2(T-1)} = \sqrt{\frac{1}{n(n-1)}} \sum_{i=1}^{n-1} \sum_{j=i+1}^n (T\hat{\rho}_{ij}^2 - 1) - \frac{n}{2(T-1)} \quad (10)$$

$$\text{where, } LM_P = \sqrt{\frac{1}{n(n-1)}} \sum_{i=1}^{n-1} \sum_{j=i+1}^n (T\hat{\rho}_{ij}^2 - 1).$$

After the examination and confirmation of CSD among the countries, it is necessary to explore the integration order of all the series depending on the second unit root generation tests, where the first-generation tests that take up the CS independence cannot be used. Therefore, we used two tests: (Pesaran, 2007) and (Herwartz and Siedenburg, 2008). Pesaran (2007) proposed a second-generation unit root test that considers the cross-sectionally augmented IPS (CIPS) in Eq. (11), which provides valid and efficient estimates while considering the cross-section dependence in the panel (Zhou et al., 2022).

$$CIPS(N, T) = N^{-1} \sum_{i=1}^N t_i^*(N, T) \quad (11)$$

Herwartz and Siedenburg (2008) test, based on the White-type covariance estimator, is determined by Eq. (12).

$$t_{HS} = \frac{\sum_{t=1}^T y'_{t-1} \Delta y_t}{\sqrt{\sum_{t=1}^T y'_{t-1} \hat{e}_t \hat{e}_t' y_{t-1}}} \quad (12)$$

The last step in this paper is to investigate the causal relationship among the variables using Dumitrescu and Hurlin (2012) test, a technique implemented for both balanced and unbalanced panel data series. Additionally, to the long-run and short-run estimations presented above, using the Dumitrescu-Hurlin test, we can find out important aspects related to the causal association between the variables. The pairwise Dumitrescu-Hurlin panel causality results emphasize the one-way or two-way causal association between the indicators, and this is critical for policymakers to embrace a correct approach in drawing and implementing policies. Dumitrescu and Hurlin (2012) provided an extension of (Granger, 1969) to detect causality in panel data. The regression is given by Eq. (13).

$$y_{i,t} = \alpha_i + \sum_{k=1}^K \gamma_{ik} y_{i,t-k} + \sum_{k=1}^K \beta_{ik} x_{i,t-k} + \varepsilon_{i,t} \quad (13)$$

where,  $i = 1, \dots, N$  and  $t = 1, \dots, T$ , and  $x_{i,t}$  and  $y_{i,t}$  are the observations of two stationary variables for individual  $i$  in period  $t$ . The lag order  $K$  is

assumed to be identical for all individuals, and the panel must be balanced (Lopez and Weber, 2017) (see Table 1).

## 5. Results and discussion

Descriptive statistics are presented in Table-2. Firstly, it is clear that the Asian sample is more polluting than the European sample, as we note that the average for emerging European countries was around 17.113, while the average for emerging Asian countries was 19.145. Second, the outcomes revealed a relative difference in the proportion of informal economy in the two samples in the range of 4.225 and 3.253. However, with a larger standard deviation in the case of Asian countries by 0.426 compared to 0.235 in European countries, this indicates a large disparity within Asian countries in the matter of informal economy. Third, the statistics revealed that the Asian sample has a significant proportion of renewable energy consumption, with an average of 3.181 in the Asian sample, compared to 2.724 in the European sample, with significant differences between the countries of both samples due to relatively large standard deviations ranging between 0.880 and 0.534, respectively. Fourth, regarding governance indicators, there is no doubt that the European sample outperforms the Asian sample in all indicators, as seen by the high averages for the four indicators. This indicates that European countries have greater and more effective governance through effective regulatory laws, more effective control of corruption, and an important role for governments in comparison to Asian countries, which have relatively weak government and control over corruption, as well as ineffective laws in most cases due to the relative spread of corruption in these countries. Before proceeding with the econometric study, it is necessary to ensure heterogeneity between the countries to avoid any inaccurate outcomes. Thus, we use Pesaran and Yamagata (2008) test and Pesaran (2007).

Furthermore, Table-3 presents the correlation matrix among the variables. The outcomes showed that EF is negatively correlated with all the independent variables except with GE in the European sample with

**Table 1**  
Variable description and data sources.

| Variable                           | Definition                                                                                                                                                                                                                          | Sources                            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Ecological footprint (EF)          | The ecological assets necessary for a population to produce the natural resources it consumes, using six indicators: carbon footprint, built-in land, forest, cropland and pasture, and fisheries measured in a consolidated metric | Global Footprint Network (2022)    |
| Informal economy (INF)             | Dynamic General Equilibrium (DGE) model-based estimates of informal output (% of official GDP)                                                                                                                                      | Elgin et al. (2021)                |
| Renewable energy consumption (REN) | Renewable energy consumption (% of total final energy consumption)                                                                                                                                                                  | World Development Indicators (WDI) |
| Government Effectiveness (GE)      | General perceptions of the quality of public services, its independence from political pressures, policy formulation and implementation, and the credibility of the government's commitment to these policies.                      | WGI (2022)                         |
| Regulatory Quality (RQ)            | General perceptions of the ability of the government to formulate and implement sound policies and regulations to promote private sector development                                                                                | WGI (2022)                         |
| Rule of Law (RL)                   | General perceptions regarding the reliability of the society rules, and the quality of contract enforcement, property rights                                                                                                        | WGI (2022)                         |
| Control of Corruption (CC)         | General perceptions of the extent to which public power is exercised for private gain                                                                                                                                               | WGI (2022)                         |

**Table-2**  
Descriptive statistics.

| European emerging countries (EEC) |        |        |        |        |        |        |        |
|-----------------------------------|--------|--------|--------|--------|--------|--------|--------|
| Variables                         | CC     | EF     | GE     | INF    | REN    | RL     | RQ     |
| Mean                              | 4.132  | 17.113 | 4.225  | 3.151  | 2.724  | 4.212  | 4.327  |
| Median                            | 4.175  | 17.091 | 4.294  | 3.192  | 2.749  | 4.251  | 4.330  |
| Max                               | 4.421  | 18.112 | 4.438  | 3.585  | 3.751  | 4.438  | 4.466  |
| Min                               | 3.707  | 16.050 | 3.773  | 2.781  | 1.532  | 3.790  | 3.989  |
| Std. Dev                          | 0.169  | 0.677  | 0.182  | 0.235  | 0.534  | 0.176  | 0.088  |
| Skewness                          | −0.463 | −0.170 | −0.991 | −0.367 | −0.036 | −0.638 | −0.981 |
| Kurtosis                          | 2.348  | 1.701  | 2.640  | 1.901  | 2.467  | 2.238  | 4.802  |
| Jarque Berra                      | 6.363  | 8.935  | 20.130 | 8.661  | 1.433  | 10.953 | 35.191 |
| Prob.                             | 0.041  | 0.011  | 0.000  | 0.013  | 0.488  | 0.004  | 0.000  |
| observations                      | 119    | 119    | 119    | 119    | 119    | 119    | 119    |
| Asian emerging countries (AEC)    |        |        |        |        |        |        |        |
| Mean                              | 3.525  | 19.145 | 3.943  | 3.253  | 3.181  | 3.714  | 3.750  |
| Median                            | 3.611  | 18.806 | 3.986  | 3.372  | 3.497  | 3.756  | 3.808  |
| Max                               | 4.209  | 21.215 | 4.453  | 3.890  | 3.955  | 4.264  | 4.336  |
| Min                               | 2.136  | 18.223 | 3.117  | 2.450  | 0.672  | 2.851  | 2.735  |
| Std. Dev                          | 0.426  | 0.837  | 0.310  | 0.426  | 0.880  | 0.357  | 0.355  |
| Skewness                          | −0.676 | 1.303  | −0.751 | −0.180 | −1.759 | −0.575 | −0.395 |
| Kurtosis                          | 3.215  | 3.407  | 3.477  | 1.727  | 4.721  | 2.336  | 2.288  |
| Jarque Berra                      | 9.296  | 34.529 | 12.344 | 8.670  | 76.057 | 8.746  | 5.620  |
| Prob.                             | 0.009  | 0.000  | 0.002  | 0.013  | 0.000  | 0.012  | 0.060  |
| observations                      | 119    | 119    | 119    | 119    | 119    | 119    | 119    |

**Table-3**  
Correlation matrix.

| European emerging countries (EEC) |                   |                   |                   |                  |                  |                  |       |
|-----------------------------------|-------------------|-------------------|-------------------|------------------|------------------|------------------|-------|
| Variables                         | EF                | INF               | REN               | GE               | RQ               | RL               | CC    |
| EF                                | 1.000             |                   |                   |                  |                  |                  |       |
| INF                               | −0.179** (0.049)  | 1.000             |                   |                  |                  |                  |       |
| REN                               | −0.469*** (0.000) | 0.438*** (0.000)  | 1.000             |                  |                  |                  |       |
| GE                                | 0.407*** (0.000)  | −0.640*** (0.000) | −0.116 (0.207)    | 1.000            |                  |                  |       |
| RQ                                | −0.088 (0.339)    | −0.639*** (0.000) | −0.202** (0.026)  | 0.733*** (0.000) | 1.000            |                  |       |
| RL                                | −0.268*** (0.003) | −0.552*** (0.000) | 0.069 (0.455)     | 0.854*** (0.000) | 0.691*** (0.000) | 1.000            |       |
| CC                                | −0.446*** (0.000) | −0.460*** (0.000) | −0.089 (0.330)    | 0.891*** (0.000) | 0.682*** (0.000) | 0.874*** (0.000) | 1.000 |
| Asian emerging countries (AEC)    |                   |                   |                   |                  |                  |                  |       |
| Variables                         | EF                | INF               | REN               | GE               | RQ               | RL               | CC    |
| EF                                | 1.000             |                   |                   |                  |                  |                  |       |
| INF                               | −0.465*** (0.000) | 1.000             |                   |                  |                  |                  |       |
| REN                               | 0.317*** (0.000)  | −0.109** (0.038)  | 1.000             |                  |                  |                  |       |
| GE                                | 0.063 (0.491)     | 0.138 (0.132)     | −0.722*** (0.000) | 1.000            |                  |                  |       |
| RQ                                | −0.129 (0.159)    | 0.415*** (0.000)  | −0.703*** (0.000) | 0.813*** (0.000) | 1.000            |                  |       |
| RL                                | 0.191** (0.036)   | 0.012 (0.889)     | −0.633*** (0.000) | 0.860*** (0.000) | 0.743*** (0.000) | 1.000            |       |
| CC                                | 0.049 (0.593)     | 0.037 (0.687)     | −0.684*** (0.000) | 0.836*** (0.000) | 0.755*** (0.000) | 0.881*** (0.000) | 1.000 |

Note: (.) denotes probabilities; \*, \*\*, \*\*\* denotes significance at 10, 5 and 1% significance level respectively.

1% significance level (SL), meaning that these indicators could decrease emissions. However, the outcomes for Asian sample revealed a negative association among INF and EF, and RQ with a −0.465 coefficient at 1% SL, respectively −129 (but not a statistically significant one) and a positive correlation with REN and RL at 5% SL. At the same time, there is no evidence of 5% SL between EF and GE, RQ, and CC in this sample.

### 5.1. Slope homogeneity test

The results presented in Table-4 confirmed the homogeneity among the counties in EEC at 5% SL according to both parameters ( $\Delta$  and  $\Delta adj$ ) that are not statistically significant at 5% SL. This result greatly illustrates the homogeneity and similarity in the environmental situation of

**Table-4**  
Pesaran and Yamagata (2008) Test results.

| Tests        | European emerging countries (EEC) |         | Asian emerging countries (AEC) |         |
|--------------|-----------------------------------|---------|--------------------------------|---------|
|              | Delta                             | P-value | Delta                          | P-value |
| $\Delta$     | −1.114                            | 0.265   | 2.594***                       | 0.009   |
| $\Delta adj$ | −1.575                            | 0.115   | 3.565***                       | 0.000   |

Note: \*\*\* denotes significance at 1% significance level.

the seven countries that comprise this group, especially in light of the tremendous structural similarity between them and the great convergence in their economic indicators. Conversely, the AEC results showed heterogeneity among the countries because the two parameters are statistically significant at 1% SL. In contrast to prior findings, it is evident that the seven Asian countries that comprised the second study sample had considerable basic disparities in climatic conditions. This is especially true given the vast disparities of economies and their economic structures, with some of them defined by a significant production base, such as India, Vietnam, and Malaysia.

#### 5.1.1. Cross-section dependence analysis

The results inspired by Table-5 revealed clearly that the null hypothesis of no cross-section dependence could not be accepted at 5% significance level in both groups. This result confirms that any shock in all variables in one country would automatically affect the other countries in the sample; this result is logical since these countries belong to the same region, meaning they have similar characteristics (economically, environmentally, and even culturally). Therefore, the following steps (unit root tests, regression estimation, and causality) should account for this issue to avoid spurious findings.

**Table-5**

Cross-sectional dependence tests results.

| Variables | European emerging countries (EEC) |           |           | Asian emerging countries (AEC) |           |           |
|-----------|-----------------------------------|-----------|-----------|--------------------------------|-----------|-----------|
|           | BP LM                             | PS LM     | BCS LM    | PB LM                          | PS LM     | BCS LM    |
| EF        | 124.402***                        | 15.955*** | 15.736*** | 229.870***                     | 32.229*** | 32.010*** |
| INF       | 314.617***                        | 45.306*** | 45.087*** | 339.927***                     | 48.285*** | 48.066*** |
| REN       | 261.218***                        | 37.066*** | 36.477*** | 141.467***                     | 18.588*** | 18.369*** |
| RQ        | 52.463***                         | 4.854***  | 4.636***  | 63.399***                      | 6.542***  | 6.323***  |
| RL        | 138.941***                        | 18.198*** | 17.980*** | 42.505***                      | 3.318***  | 3.099***  |
| CC        | 76.653***                         | 8.587***  | 8.368***  | 49.415***                      | 4.384***  | 4.165***  |
| GE        | 35.836**                          | 2.289**   | 2.070**   | 54.261***                      | 5.132***  | 4.913***  |

Note: \*, \*\*, \*\*\* denotes significance at 10, 5, and 1% significance level respectively.

## 6. Unit root tests

The results are presented in Table-6. On the one hand, Pesaran (2007) CIPS test showed that REN, RL, CC, and GE are I(0) variables in EEC since we cannot reject the null hypothesis of no unit root in this series at 5% SL. However, we cannot accept the null hypothesis for EF, INF, and RQ, where the series becomes stationary after the first difference. Accordingly, these three variables are I(1) series. In the same test with AEC, the outcomes revealed that all the variables are I(1) series since they become stationary after taking the first differences except for RL, which turned out to be I(0). On other hand, Herwartz and Siedenburg (2008) test concluded that all the variables are I(1) series both in EEC and AEC except GE in EEC, which has been confirmed to be I(0). Overall, the outcomes of the two tests concluded that most of the variables under consideration are I(1) series without any I(2) series; this result is ideal for using the Pooled Mean Group Autoregressive Distributed Lag (PMG-ARDL) estimation in order to estimate the long run and short run effects of informality, institutional quality and renewable energy consumption on ecological footprint.

## 7. PMG-ARDL estimations

After passing the pre-tests, such as homogeneity, cross-section, and unit root tests, we concluded that the PMG-ARDL model is the best model for our data. Accordingly, the results presented in Table-7 showed several outcomes as follows. First, the long-run estimation in EEC (Figure-1) showed a negative significant effect from REN, RL, and GE to EF. Hence, this outcome implies that any increase in renewable energy consumption, rule of law, and government effectiveness by 10% will automatically decrease ecological footprints by 0.42%, 10.61%, and

**Table-6**

Unit root analysis.

| Variables                           | European emerging countries (EEC) |                | Asian emerging countries (AEC) |                |
|-------------------------------------|-----------------------------------|----------------|--------------------------------|----------------|
|                                     | Level                             | 1st difference | Level                          | 1st difference |
| Pesaran CIPS (2007) test            |                                   |                |                                |                |
| EF                                  | -1.711                            | -4.680***      | -1.270                         | -3.726***      |
| INF                                 | -0.872                            | -2.833***      | -1.046                         | -2.850***      |
| REN                                 | -2.687**                          | -4.348***      | -1.249                         | -2.992***      |
| RQ                                  | -1.860                            | -4.399***      | -2.069                         | -4.182***      |
| RL                                  | -2.812***                         | -5.195***      | -2.919***                      | -3.946***      |
| CC                                  | -2.414**                          | -3.990***      | -2.133                         | -4.007***      |
| GE                                  | -2.419**                          | -4.336***      | -1.870                         | -4.068***      |
| Herwartz and Siedenburg (2008) test |                                   |                |                                |                |
| EF                                  | -1.393                            | -2.342***      | -0.217                         | -1.873**       |
| INF                                 | -0.272                            | -2.427***      | -0.215                         | -1.795**       |
| REN                                 | 0.371                             | -2.130**       | 1.364                          | -1.725**       |
| RQ                                  | -0.598                            | -2.125**       | 0.136                          | -1.816**       |
| RL                                  | 0.576                             | -1.888**       | -0.651                         | -2.312***      |
| CC                                  | -0.781                            | -2.142**       | -0.863                         | -1.902**       |
| GE                                  | -1.952**                          | -2.492***      | -0.472                         | -2.445***      |

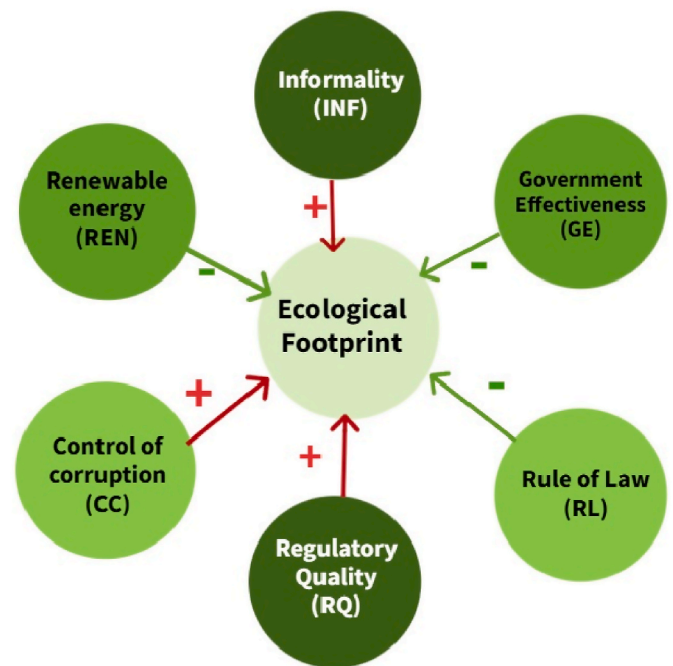
Note: \*, \*\*, \*\*\* denotes significance at 10, 5, and 1% significance level respectively. Critical values at 10, 5, and 1% significance level are: 2.21; -2.34, and -2.60, respectively.

**Table 7**

PMG-ARDL estimation results.

| Variables            | EEC         |             | AEC         |             |
|----------------------|-------------|-------------|-------------|-------------|
|                      | Coefficient | Probability | Coefficient | Probability |
| Long run estimation  |             |             |             |             |
| INF                  | 1.276***    | 0.000       | -3.577***   | 0.000       |
| REN                  | -0.042***   | 0.000       | -0.085      | 0.234       |
| CC                   | 0.506***    | 0.000       | -0.180***   | 0.000       |
| RQ                   | 1.254***    | 0.000       | 0.404***    | 0.000       |
| RL                   | -1.610***   | 0.000       | -0.258***   | 0.000       |
| GE                   | -0.995***   | 0.000       | -0.244***   | 0.000       |
| Short run estimation |             |             |             |             |
| ECT                  | -0.719***   | 0.000       | -0.718**    | 0.041       |
| D(INF)               | 2.335       | 0.562       | 5.129**     | 0.047       |
| D(REN)               | -0.106      | 0.456       | -0.181      | 0.603       |
| D(CC)                | 0.207       | 0.485       | 0.037       | 0.273       |
| D(RQ)                | -0.331      | 0.508       | -0.042      | 0.605       |
| D(RL)                | 0.550       | 0.144       | 0.380**     | 0.049       |
| D(GE)                | 0.222       | 0.472       | 0.109       | 0.493       |
| Constant             | 11.815***   | 0.000       | 23.479**    | 0.042       |

Note: D denotes differences operator; \*, \*\*, \*\*\* denotes significance at 10, 5 and 1% significance level respectively and ECT represents Error Correction Term.

**Figure-1.** Long-term estimations for EEC

9.99%, respectively; this result shows the importance of these variables in improving environmental quality in these countries. Contrariwise, RQ and CC coefficients, in contrast to RL and GE, were significantly positive,

meaning that regulatory quality and control of corruption could contribute to environmental deterioration, where any increase in RQ and CC by 10% will increase EF by 12.54% and 5.06% in the long run respectively. Furthermore, the findings exposed that INF contributes to an increase in EF at 1% SL, where any rise in INF by 10% will upsurge the emissions by 12.76%. Thus, this finding revealed the harmful effect of the informal economy on environment quality, being in line with previous findings (Chen et al., 2018; Ortiz et al., 2022; Pang et al., 2021; Sultana et al., 2022a), but different from Benkraiem et al. (2019) outcomes. Given these outcomes, policymakers from European emerging countries should proceed to reduce informality to better address net-zero emissions targets.

Conversely, in the short-run estimation, it is clear that the error correct term (ECT) is between 0 and -1 and is significant at 1% SL. This result confirms the existence of a long-term relationship between the variables under study in the presence of a structural modification of the system after any possible shock, where the coefficient shows that this system corrects 71.9% of its imbalances after each shock to return to the equilibrium position, which means returning to the equilibrium state after 1.5 years. Passing to the AEC sample, the main result is that INF plays a vital role in environmental quality in such nations since any increase in INF by 10% will alleviate the ecological footprint 35.77% in the long run (Figure-2). Accordingly, this result explains the significant impact of informal economy on the environmental situation, especially when we note that its impact is double that achieved in the European sample. This fact suggests that informal activities are practiced in sectors which are not energy intensive and not harmful to environment, confirming the results of Benkraiem et al. (2019), Nguyen and Nguyen (2023). Besides, the outcomes confirmed the positive impact of RQ on EF, indicating that an increase in the regulatory quality contributes to emissions intensifications. However, the other governance indicators, such as CC, RL, and GE, could provide an ideal solution to reduce emissions and improve environmental conditions by achieving the opposite effect of these variables on EF. The increase in CC, RL, and GE by 1% will shrink the EF by 0.18%, 0.25%, and 0.24%, respectively. Nevertheless, the results showed that there is not enough evidence at 5% SL that REN can affect emissions, and this may be mainly due to the low use of renewable energies in these countries and their weak ratio to the total percentage of energy consumed.

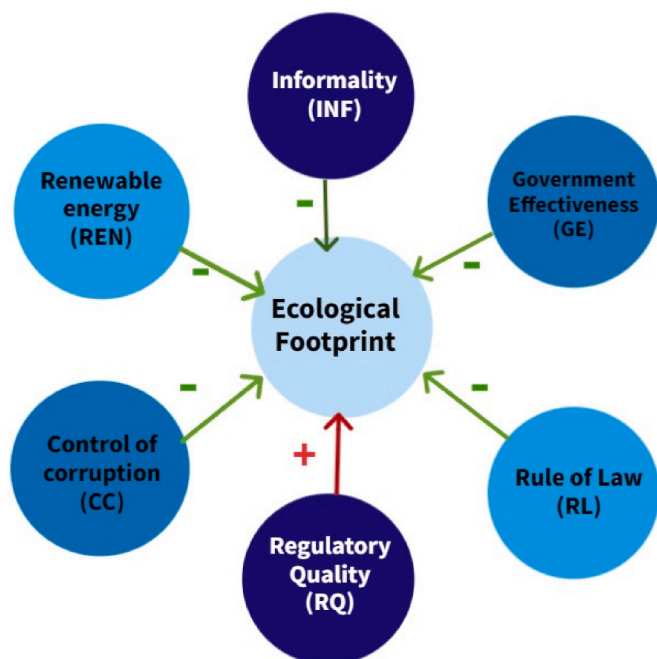


Figure-2. Long-term estimation for AEC

Shifting focus to the immediate results, the outcomes confirmed the existence of the adjustment after any shock in the system back to the equilibrium state; after any shock affecting the relationship between these variables, the system will correct 71.9% each year to return to the equilibrium state, which means achieving this in more than 1.5 years. Furthermore, the results show a significant effect of INF on EF. In this sense, the informal economy contributes to the promotion of environmental degradation in the short term, playing a different role than on long run, confirming the results of Ortiz et al. (2022) and Sultana et al., 2022a. Similarly, the findings revealed that RL could also increase emissions in the short run since the occurrence of a positive impact on EF at 5% SL.

## 8. Causal relationship

The findings are presented in Table-8. For EEC countries, the outcomes revealed the existence of a bidirectional causality between EF and INF, meaning that both variables are very important in predicting each other. Moreover, there is unidirectional causality running from REN to EF, REN to RQ, REN to RL, INF to RQ, and INF to RL (Figure-3). On the other hand, similar to EEC, the findings indicated a bidirectional causal relationship between INF and EF in AEC, in addition to bidirectional causality between CC and INF and between RL and INF. Besides, we find unidirectional causality from INF to RQ, RL to REN, and RQ to RL (Figure-4). Both countries group form this study should synergistically set political targets to cope with these phenomena in their development path.

## 9. Conclusion and policy implications

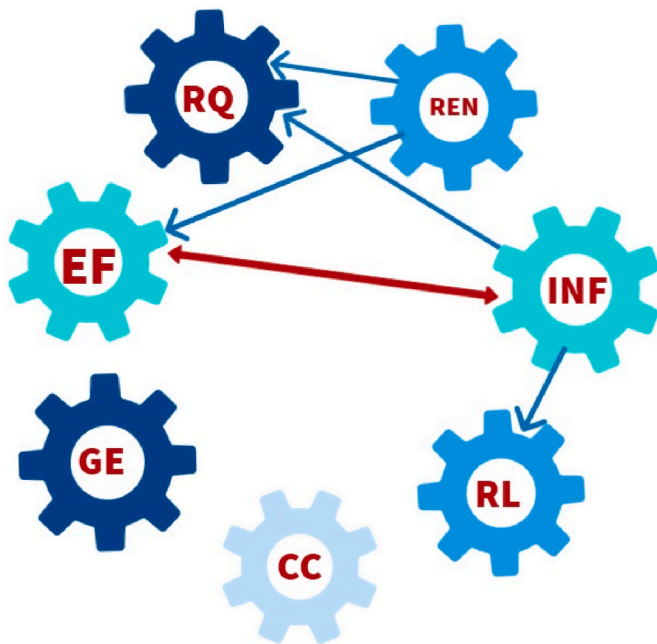
This paper was to investigate the effect of informal economy, governance indicators, and renewable energy consumption on environment quality in selected European and Asian emerging countries over the period 2002–2018. In order to achieve this goal, we used different tests and estimation procedures such as slope homogeneity, cross-section dependence, and unit root procedures as primary tests. In addition, we used the PMG-ARDL approach to detect both samples' long-run and short-run effects. Finally, we used the Dumetriscu-Hurlin Granger causality test to explore the causal relationship among the variables. Accordingly, we used different variables for our series. First, we employ ecological footprints as an environmental variable superior to CO<sub>2</sub> emissions, which solely focus on air pollution, in contrast to EF, which focus on air, water, and soil. Second, we use the share of the informal economy output as an indicator of informality in these countries. Third, the paper analyzed four governance indicators: Government Effectiveness, Regulatory Quality, Rule of Law and Control of Corruption to get a clear idea about the impact of governance on environmental deterioration. Lastly, renewable energy consumption was included as a control variable to investigate the role of this kind of energy in climate change mitigation in both samples. This study confirmed the homogeneity among the EEC countries EEC, but not in AEC countries. This means a greater similarity in the environmental aspect of the European sample in contrast to the Asian sample. Second, both samples showed cross-section dependence, so any shock in one of the variables in any country would be transmitted to the other countries in both samples. More importantly, the findings showed different effects of informality in the long run in EEC from AEC, validating the heterogeneity of the results. The outcomes highlight the positive effect of informality in the EEC sample; the shadow economy increases pollution and pulls back the policymakers' efforts to mitigate environmental risks. Contrary-wise, in the AEC sample, the harmful effect of informality on environmental quality is observed only in the short run; in the long run, the positive impact on the income of people that are working in non-statutory conditions may alleviate the negative impact on environment. As Bolarinwa and Simatele (2023) stated, a particular informality level may be expedient in the feedback effect framework, as there is a double



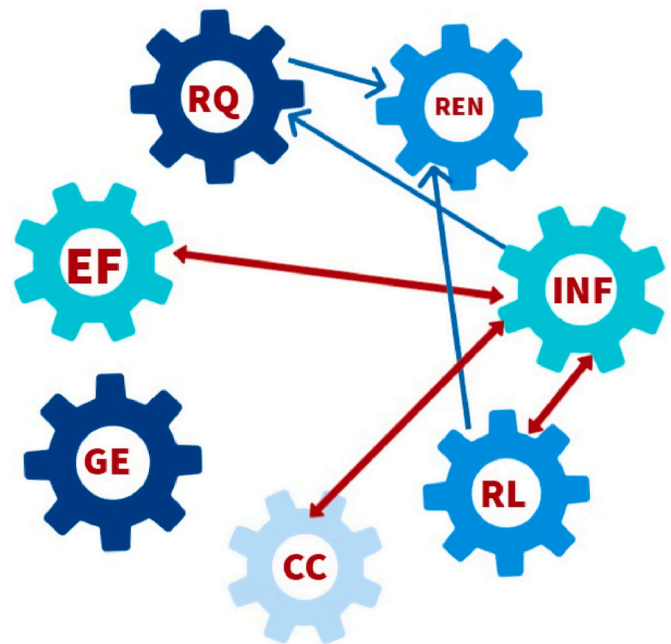
**Table-8**  
Dumitrescu-Hurlin causality results.

| European emerging countries (EEC) |           |           |           |       |           |          |          |
|-----------------------------------|-----------|-----------|-----------|-------|-----------|----------|----------|
| Impulse variable                  |           |           |           |       |           |          |          |
| Variables                         | EF        | INF       | REN       | GE    | RQ        | RL       | CC       |
| EF                                | –         | 10.804*** | 8.761***  | 2.307 | 1.126     | 2.357    | 3.170    |
| INF                               | 13.399*** | –         | 4.714*    | 0.940 | 3.977     | 3.050    | 3.289    |
| REN                               | 1.554     | 8.531     | –         | 3.042 | 2.440     | 2.092    | 1.342    |
| GE                                | 4.033     | 2.694     | 2.923     | –     | 4.565*    | 3.020    | 2.701    |
| RQ                                | 3.199     | 5.190**   | 6.249***  | 3.159 | –         | 4.082    | 4.519*   |
| RL                                | 3.094     | 5.638**   | 13.800*** | 2.040 | 1.255     | –        | 5.370**  |
| CC                                | 2.045     | 2.796     | 2.038     | 1.849 | 1.418     | 1.553    | –        |
| Asian emerging countries (AEC)    |           |           |           |       |           |          |          |
| Variables                         | EF        | INF       | REN       | GE    | RQ        | RL       | CC       |
| EF                                | –         | 8.252***  | 1.216     | 2.458 | 2.923     | 3.077    | 2.732    |
| INF                               | 6.340***  | –         | 4.178     | 1.801 | 4.763*    | 5.013**  | 5.313*** |
| REN                               | 2.439     | 4.279     | –         | 3.820 | 11.351*** | 6.844*** | 2.737    |
| GE                                | 4.879*    | 2.015     | 3.673     | –     | 3.784     | 2.032    | 3.657    |
| RQ                                | 1.570     | 4.908**   | 1.851     | 3.586 | –         | 3.745    | 3.833    |
| RL                                | 4.719*    | 5.416**   | 4.432     | 3.470 | 4.916**   | –        | 2.276    |
| CC                                | 4.555*    | 7.188***  | 3.619     | 3.753 | 1.745     | 1.603    | –        |

Note: \*, \*\*, \*\*\* denotes significance at 10, 5 and 1% significance level respectively.



**Figure-3.** Causality in EEC



**Figure 4.** Causality in AEC

causality between informality and ecological footprint in both analyzed sample. Fourth, the outcomes revealed that RL and GE could be essential in reducing EF in the long run in both regions.

On other hand, it is clear from the results that RQ could deteriorate environment in EEC and AEC, noting that the inverse effect of RL is relatively greater than the direct effect of RQ. Fifth, the results showed that CC contributes to increasing emissions in EEC, which means that it has an adverse effect on environment but could be a good option to diminish the negative impact on environmental quality in AEC. Sixth, it was found in both samples that the consumption of renewable energies contributes significantly to improving environmental conditions, with a more significant contribution in the European sample. Our outcomes of study show the important impact of informality and institutional quality on environmental conditions. However, the effects differ in sign and intensity from one group to another, highlighting informality's negative potential on the European sample's ecological footprint. This should be a caveat for policymakers to take emergent measures to reduce it and

transfer these activities into the formal economy, which will improve working conditions and social benefits for the employees and increase the fiscal space (Heller, 2005) for new environmentally friendly investments and projects.

Moreover, based on these results, we conclude that governance variables play a critical and significant role in influencing various economic activities, such as investment and consumption; and these put additional pressure on energy consumption. Therefore, governments are strongly required to control all the circumstances surrounding economic activity through clear laws and an influential role in encouraging institutions and individuals to seek to improve environmental conditions. For example, it is possible to tax only activities that are damaging environment in order to decrease fossil fuel consumption and stimulate renewable energy. Furthermore, in both samples, it is critical to significantly encourage the use of renewable energy sources, such as wind, solar, water, and nuclear energies, because these countries have significant capabilities to increase their share of renewable in total energy

consumption. Likewise, increased government spending on research and development to improve technological capabilities in the renewable energy sector is strongly recommended. On other hand, opening the door to direct foreign investment might be a significant chance to bring current environmentally friendly technologies to build infrastructure and simplify the energy transition. Informality must be evaluated, and policymakers should adopt better regulations to increase the insertion of informal workers. This will create a compliance framework for women and low-skilled young people, predominant in the informal economy. In this sense, policymakers should act to provide more opportunities in the formal sectors for all the actors currently working in the informal sector. In some cases, workers in the informal sector earn more than those in the formal sector. However, individuals may choose to participate in the informal economy due to a lack of access to financial resources or skills to compete in formal economy. Sometimes, they complement their formal-driven income. On other hand, many workers in the informal sector manage only to secure their basic needs, not being able to work in the formal framework and without benefiting from social protection. A crucial aspect that must be considered and which is also emphasized by the study results, especially in the case of Asian Emerging countries, is the fact that the same bidirectional causality is also found between the institutional variables (rule of law, control of corruption) and informality. So, it is evident that in this geographical area, public institutions must be more robust and more relevant, even if the effect on ecological footprint does not have a clear negative dimension, but the economic effect is important and must be prioritized.

This study has disclosed that informality has diversified effects on ecological footprint in the two selected groups, which exhibits different perspectives regarding political options. Emerging European countries should focus on measures to inhibit informality to minimize its negative impact on the quality of environment. On other hand, emerging countries in Asia should center their action on enhancing institutional quality. Thus, they can support the vulnerable to a smooth transition to formality, allowing them to earn the necessary income. It is vital for these countries to corroborate targets in achieving sustainable development.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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